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BOLT ASSEMBLY FOR FIREARM

Abstract

A modular two-piece bolt system for firearms with blowback actions provides a rear bolt piece which detachably couples to plural different front bolt pieces each of different mass. The front bolt pieces have a similar configuration and common coupling interface which forms an interlocked relationship with a complementary configured coupling interface on the rear bolt piece to prevent axial separation of the pieces when the action of the firearm is cycled. The system allows various front bolt pieces to be selected and used interchangeably with the rear bolt piece so that a front bolt piece of proper mass can be matched to the energy of ammunition to be fired. The front and rear bolt pieces when coupled define a continuous extended longitudinal passage which receives the rear breech end portion of the firearm barrel therethrough to reduce the projected length of the barrel.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of U.S. patent application Ser. No. 18/238,235 filed Aug. 25, 2023, which claims the benefit of priority to U.S. Provisional Patent Application No. 63/498,124 filed Apr. 25, 2023, and U.S. Provisional Patent Application No. 63/400,939 filed Aug. 25, 2022; the entireties of the foregoing applications are all incorporated herein by reference.

BACKGROUND

[0002] The present invention generally relates to firearms, and more particularly to bolts for blowback operated firearms such as long guns including rifles, carbines, and shotguns.

[0003] The mass of bolts used in auto-loading firearms having a blowback type operating system to cycle the action between open and closed breech positions must be precisely selected. In blowback actions, the mass of the slideable bolt carried by the receiver and the recoil spring maintains a closed breech until the firearm is fired. Accordingly, the masses of such bolts must be matched to the propellant powder charge or load of the particular cartridge to be fired in the firearm. Therefore, multiple bolts with different masses must be fabricated by the firearm manufacturer. In addition, if a user decides to change the caliber of their firearm, a complete bolt must be purchased which can be costly.

[0004] Improvement in bolts for firearms with blowback type operating systems is desired which have some degree of modularity to provide bolt fabrication and replacement flexibility and economies to accommodate firing different ammunition with different powder charges or loads in the firearm.

SUMMARY OF THE INVENTION

[0005] According to one embodiment, a modular two-piece bolt for a blowback action may be provided forming a bolt assembly including a front bolt piece detachably coupled to a rear bolt piece. The rear bolt piece defines the breech face which is movable into and out of battery (i.e. engagement) with the rear breech end of the barrel via cycling the bolt assembly. Advantageously, the interchangeable front bolt piece may have a mass selected to provide the necessary total mass of the bolt assembly (front and rear bolt pieces) to match the propellant powder load or charge of the ammunition cartridge for which the firearm is chambered. This advantageously provides bolt fabrication and user replacement economies.

[0006] A modular bolt system according to the present disclosure comprises a plurality of front bolt pieces of similar configuration but each having a different mass. The front bolt pieces all share a common coupling interface configured for detachable coupling to the mating coupling feature of the rear bolt piece. The coupling feature of the rear bolt piece therefore has a configuration compatible for use with the plurality of front bolt pieces via the common coupling interface. This allows a standardized design of a rear bolt piece to be fabricated by the firearm manufacturer which is interchangeable with the different front bolt pieces of varying mass.

[0007] Certain calibers of cartridges may be commercially offered in a variety of different propellant powder charges or loads. In such cases, the same rear bolt piece may be used which

comprises the breech face that is compatible for use with all of the cartridges of the same caliber but with different charges in each. An appropriate front bolt piece may be selected and coupled to the rear bolt piece which gives the entire bolt assembly the total mass required for the powder charge of the cartridge to be fired. In this situation, the rear bolt piece therefore does not need to be replaced when selecting the appropriate front bolt piece.

[0008] The firearm in one embodiment may comprise a bolt-over-barrel design in one embodiment. Each of the front and rear bolt pieces defines a portion of an extended longitudinal passage which receives a length of the rear breech end portion of the firearm barrel. The rear breech end portion of the barrel therefore extends a substantial distance into the axial cavity of the receiver which slideably receives the bolt assembly, as further described herein. This advantageously shortens the forward projected length of the barrel which is therefore amendable for use in a carbine firearm in contrast to a standard rifle having a longer barrel projected length. The two piece bolt assembly allows for assembly and disassembly of the firearm while having the rear of the receiver immovable. Otherwise, the length of the bolt assembly would be greater than the opening it is allowed to be inserted into.

[0009] In one aspect, a firearm with bolt assembly for a blowback operating system comprises: a longitudinal axis; a receiver defining an axially elongated internal cavity; a barrel supported by the receiver, the barrel defining an axial bore extending longitudinally from a rear breech end to a front muzzle end of the barrel; a bolt assembly slideably disposed in the cavity of the receiver, the bolt assembly axially movable between a forward closed breech position in battery with the rear breech end of the barrel and a rearward open breech position; a recoil spring biasing the bolt forward towards the closed breech position; the bolt assembly comprising a front bolt piece detachably coupled to a rear bolt piece; the front and rear bolt pieces each comprising a longitudinal passage coaxial with the longitudinal axis of the firearm, the longitudinal passage of the front bolt piece in communication with the longitudinal passage of the rear bolt piece to collectively form a continuous extended longitudinal passage; wherein a rear breech end of the barrel is disposed inside the extended longitudinal passage. The front and rear bolt pieces are keyed into one another by a coupling protrusion disposed on the front or rear bolt piece interlocked with a mating coupling recess disposed on the other of the front or rear bolt piece.

[0010] According to another aspect, a bolt assembly for a firearm with blowback operating system comprises: a longitudinal axis; an axially elongated front bolt piece internally comprising a first longitudinal passage coaxial with the longitudinal axis, the first longitudinal passage being configured to receive a barrel of the firearm; an axially elongated rear bolt piece internally comprising a second longitudinal passage coaxial with the longitudinal axis, the second longitudinal passage being configured to receive the barrel of the firearm; the second longitudinal passage being in communication with the first longitudinal passage of the front bolt piece to collectively form a continuous extended longitudinal passage; the front bolt piece being detachably coupled to the rear bolt piece and interlocked together to prevent separation in a longitudinal direction; and a breech face formed in the second longitudinal passage of the rear bolt piece.

[0011] According to another aspect, a modular bolt assembly system for a firearm with blowback operating system comprises: a longitudinal axis; a first front bolt piece having a first mass and internally comprising a first longitudinal passage coaxial with the longitudinal axis, the first longitudinal passage being configured to receive a rear portion of a first firearm barrel chambered for a first cartridge; a second front bolt piece having a second mass different than the first mass, the second front bolt piece internally comprising a second longitudinal passage coaxial with the longitudinal axis, the second longitudinal passage being configured to receive a rear portion of a second firearm barrel chambered for a second cartridge having a propellant powder charge with energy different than the first cartridge; a rear bolt piece internally comprising a third longitudinal passage coaxial with the longitudinal axis, the third longitudinal passage comprising a breech face and being configured to receive the rear portions of the first or second firearm barrel; wherein the

first and second front bolt pieces each have a common first coupling interface configured to mate with a corresponding complementary configured second coupling interface formed on the rear bolt piece; wherein the rear bolt piece is detachably coupleable to the first or second front bolt piece to form a mechanical interlock which prevents separation thereof in a direction along the longitudinal axis.

[0012] According to another aspect, a method for matching a bolt assembly to a firearm comprises: providing a firearm comprising a longitudinal axis and a receiver defining an axially extending cavity; providing a barrel chambered for a first cartridge having a first propellant powder charge; providing a rear bolt piece comprising a longitudinal passage and a breech face formed in the longitudinal passage; selecting a front bolt piece from a plurality of front bolt pieces each having a different mass and a longitudinal passage, the selected front bolt piece having a mass compatible for use with firing the first cartridge in the firearm; coupling the first front bolt piece to the rear bolt piece to form a bolt assembly; and inserting the bolt assembly into the cavity of the receiver.

[0013] Embodiments of the present invention may also provide an ambidextrous bolt release mechanism for a firearm, which is usable with the foregoing two-piece bolt. The bolt-catch release mechanism comprises a bolt catch including an axially elongated bolt catch lever pivotably mounted on one side of the firearm, and a spring-biased actuator plunger accessible from the opposite side of the firearm which cooperates with the lever to form an ambidextrous mechanism. The bolt catch lever is configured and operable to engage and lock the axially movable bolt of the firearm action in a rearward open breech position when the lever is engaged with the bolt. The manually operated plunger is transversely oriented to the bolt catch lever and slideably engageable with the lever to cause rotation thereof for disengaging and releasing the bolt from the rearward open breech position. Alternatively, the bolt catch lever may also be manually actuated directly by the user to release the bolt.

[0014] In one unique aspect of the present invention, the bolt catch lever in one embodiment includes an inclined camming ramp which forms an operating surface that is in turn acted upon by a complementary angled inclined camming surface defined by a working end of the actuator plunger opposite the operating end which a user depresses inward to actuate the plunger. A flat-to-flat sliding interface is formed in one embodiment between the inclined camming surface and ramp. The camming surface may be defined by a conical tip of the plunger in one implementation.

[0015] When the bolt is held and locked in rearward in its open breech position via the bolt catch lever, the inclined camming surface of the actuator plunger acts as a wedge that pries and rotates the bolt catch lever out of engagement with the bolt when the user-accessible operating end of the plunger is pressed inwards to slide the plunger towards the lever. This releases the bolt forward to reclose the breech.

[0016] The actuator plunger advantageously permits the user to release the bolt from both sides of the firearm; either by manually moving the integrated bolt catch lever directly on one side of the firearm, or via manually actuating the spring-loaded plunger from the other side. The spring-loaded plunger may be located on the firearm in a position selected for easy actuation by the user's index finger in one embodiment. This allows an ergonomic and natural motion for the user to increase the speed and ease of reloading the firearm, which is particularly beneficial for purposes such as competition shooting. Additionally, the spring-biased plunger is useful for shooters with an opposite dominant hand, which could cause difficulty in accessing the traditional integrated bolt catch lever on one side. These shooters may instead press the plunger inwards for actuation with their non-dominant hand after using this hand to insert the ammunition magazine, thereby permitting a speedy and easy reloading process.

[0017] The plunger may be spring-biased in a direction away from the bolt catch lever to the degree that avoids actuation of the lever until the plunger is manually actuated by the user. As a result, the plunger does not hinder mechanical function or movement of the bolt catch with unnecessary added inertial load, friction, or other encumbrances.

[0018] The firearm may comprise a blowback action in which the mass of the slideable bolt carried by the receiver and the recoil spring maintains a closed breech until the firearm is fired (or bolt is manually retracted). The blowback action bolt is not rotatable or configured to form a locked breech since the bolt does not contain radial bolt lugs which lock up with corresponding barrel locking lugs in the rear breech end of the barrel to lock the breech as in some firearms; this latter arrangement being typically used with cylindrical bolts that are rotatable between locked and unlocked position. The bolt body of the present bolt design may have a substantially rectangular cuboid configuration in some embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] The features of the exemplary embodiments will be described with reference to the following drawings where like elements are labeled similarly, and in which:

[0020] FIG. 1 is a top perspective view of a firearm with bolt catch and bolt assembly according to the present disclosure;

[0021] FIG. 2 is a left side cross sectional view thereof;

[0022] FIG. 3 is an enlarged detail taken from FIG. 2;

[0023] FIG. 4 is an enlarged detail taken from FIG. 1;

[0024] FIG. 5 is a left side rear perspective view of the mid-stock are of the firearm;

[0025] FIG. 6 is a left side view thereof;

[0026] FIG. 7 is a right side view thereof;

[0027] FIG. 8A is left side view of part of the firearm showing the bolt in a rearward open breech position and the bolt catch lever of the bolt catch engaged with the bolt to retain the open breech position;

[0028] FIG. 8B is an enlarged detail taken from FIG. 8A;

[0029] FIG. 9A is left side view of part of the firearm showing the bolt returning forward to the closed breech position and the bolt catch lever disengaged from and releasing the bolt;

[0030] FIG. 9B is an enlarged detail taken from FIG. 9A;

[0031] FIG. 10 is a transverse cross sectional view taken from FIG. 8A showing the actuator plunger of the bolt catch in an unactuated position which does not activate the bolt catch lever;

[0032] FIG. 11 is a transverse cross sectional view taken from FIG. 9A showing the actuator plunger of the bolt catch in an actuated position slideably engaged with and activating the bolt catch lever to cause rotation thereof to disengage and release the bolt;

[0033] FIG. 12 is a top exploded perspective view of the bolt, fire control insert, and pistol grip frame of the firearm;

[0034] FIG. 13 is a bottom exploded perspective view of the bolt;

[0035] FIG. 14 is a top exploded perspective view of the fire control insert with firing mechanism components and the bolt catch mounted thereon;

[0036] FIG. 15 is a bottom exploded perspective view thereof;

[0037] FIG. 16 is a top front assembled perspective view thereof;

[0038] FIG. 17 is a first top rear assembled perspective view thereof;

[0039] FIG. 18 is a second top rear assembled perspective view thereof;

[0040] FIG. 19 is a bottom assembled perspective view thereof;

[0041] FIG. 20 is a left side view of the fire control insert;

[0042] FIG. 21 is a right side view thereof;

[0043] FIG. 22 is a top view thereof;

[0044] FIG. 23 is a bottom view thereof;

[0045] FIG. 24 is a first exploded perspective view of the bolt catch comprising the bolt catch lever

and actuator plunger;
[0046] FIG. **25** is a second exploded perspective view thereof;
[0047] FIG. **26** is a rear view thereof in an assembled condition;
[0048] FIG. **27** is a front view thereof;
[0049] FIG. **28** is a top perspective view thereof;
[0050] FIG. **29** is a bottom perspective view;
[0051] FIG. **30** is a third exploded perspective view thereof;
[0052] FIG. **31** is a fourth exploded perspective view thereof;
[0053] FIG. **32** is a top view thereof in an assembled condition; and
[0054] FIG. **33** is a bottom view thereof;
[0055] FIG. **34** is first front perspective view of an alternative embodiment of a bolt assembly for the firearm;
[0056] FIG. **35** is a second front perspective view thereof;
[0057] FIG. **36** is a first bottom rear perspective view thereof;
[0058] FIG. **37** is a second bottom rear perspective view thereof;
[0059] FIG. **38** is a first transverse cross-sectional view taken from FIG. **36** through the front bolt piece;
[0060] FIG. **39** is a second transverse cross sectional view taken from FIG. **37** through the rear bolt piece;
[0061] FIG. **40** is a front perspective view of the front bolt piece of the bolt assembly of FIG. **34**;
[0062] FIG. **41** is a rear perspective view thereof;
[0063] FIG. **42** is a front perspective view of the rear bolt piece of the bolt assembly of FIG. **34**;
and
[0064] FIG. **43** is a rear perspective view thereof.

[0065] All drawings are schematic and not necessarily to scale. Parts shown and/or given a reference numerical designation in one figure may be considered to be the same parts where they appear in other figures without a numerical designation for brevity unless specifically labeled with a different part number and described herein. A reference herein to a figure by a whole number which includes multiple figures sharing the same whole number but with different alphabetical suffixes shall be construed as a general reference to all of those figures unless expressly noted otherwise.

DETAILED DESCRIPTION OF EMBODIMENTS

[0066] The features and benefits of the invention are illustrated and described herein by reference to preferred but non-limiting exemplary (“example”) embodiments. This description of the embodiments is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. Accordingly, the invention expressly should not be limited to such embodiments illustrating some possible non-limiting combination of features that may exist alone or in other combinations of features; the scope of the invention being defined by the claims appended hereto.

[0067] In the description of embodiments disclosed herein, any reference to direction or orientation is merely intended for convenience of description and is not intended in any way to limit the scope of the present invention. Relative terms such as “lower,” “upper,” “horizontal,” “vertical,” “above,” “below,” “up,” “down,” “top” and “bottom” as well as derivative thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description only and do not require that the apparatus be constructed or operated in a particular orientation. Terms such as “attached,” “affixed,” “connected,” “coupled,” “interconnected,” and similar refer to a relationship wherein structures may be secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise.

[0068] FIGS. 1-33 generally depict a magazine-fed autoloading semi-automatic firearm **20** including a bolt release mechanism and various aspects thereof according to the present disclosure. In one non-limiting embodiment, the firearm may be a long gun such as a rifle, carbine, or shotgun. The non-limiting illustrated embodiment, selected for convenience of description only as one representative firearm, shows a carbine which is essentially a rifle with space-efficient barrel projection useful for many purposes such as competitive action shooting. In other embodiments, however, the bolt release mechanism disclosed herein may be applied to pistols to lock the axially movable slide in a rearward open breech position. Accordingly, the bolt catch is not limited in its applicability to any particular firearm format alone in which it is desirable to lock the bolt or slide in a rearward open breech position either manually or automatically via operation of the magazine follower, and to release the bolt forward to reclose the breech.

[0069] Firearm **20** includes a longitudinal axis LA, receiver **21**, barrel **22** coupled thereto, bolt **50**, and a trigger-actuated firing mechanism **23** which may be supported directly by the receiver or alternatively a detachable fire control insert **23-1** as illustrated herein. Firing mechanism **23** includes movable trigger **24** for actuating the mechanism to discharge the firearm. The firearm includes a chassis or stock **80** including buttstock **81** for shouldering the firearm and mid-stock **82** to which the receiver **21** is detachably coupled by any suitable method used in the art such as via mounting pins, threaded fasteners, interlocking surfaces or protrusions, combinations thereof, or other mechanical fastening means. In one embodiment, the mid-stock may be a downwardly extending pistol grip frame **82-1** which defines a magazine well **32**. In this case, receiver **21** may be detachably mounted to the pistol grip frame **82-1**. Pistol grip frame **82-1** may include a open loop-shaped bottom trigger guard **24-1** which surrounds and helps protect the trigger **24** against unintentional actuation.

[0070] Downwardly and upwardly open magazine well **32** detachably holds ammunition magazine **33** comprising an internal magazine cavity **33-4** containing a plurality of cartridges (rounds) mounted therein. Such magazines may have a straight or curved box-type in some embodiments which contains a spring-biased stack of ammunition cartridges which are uploaded into the breech area **34** by a spring feed mechanism for chambering into the rearwardly open chamber **36** of barrel **22** by the bolt **50** in a conventional manner when cycling the action (example of box-type magazine illustrated in the figures). The spring feed mechanism disposed in magazine cavity **33-4** includes a follower **33-1** which supports a vertical stack of cartridges C and magazine spring **33-2** which acts on the follower to bias the cartridges upwards towards the open top of the magazine for feeding into the breech area of the receiver in a conventional manner (see, e.g., FIG. 3). Spring **33-2** acts between magazine floor plate **33-3** coupled to the bottom of the magazine tube and the bottom of follower **33-1** at top. The follower is operably interfaced with the bolt catch **100** to automatically actuate the catch to lock the bolt rearward as a last round hold open feature, further described herein. Any type of cartridges may be used depending on the type of cartridge for which the firearm is chambered, including centerfire or rimfire cartridges both of which are well known in the art without further undue elaboration.

[0071] The magazine **33** is removably retained in the magazine well **32** by a movable magazine latch **35**. Latch **35** may be slideably mounted to pistol grip frame **82-1** of the mid-stock **82** in one embodiment and is configured to alternately lock the magazine in the firearm or release the magazine when the latch is manually actuated by the user. Latch **35** may be mounted in the front of pistol grip frame **82-1** proximate to the front of the magazine well **32** in one embodiment as shown for front latching type magazines. However, latch **35** may alternatively be mounted on either side of the pistol grip frame **82-1** or at the rear thereof if other latching type magazines are used. The location of the latch does not limit the present invention in any manner.

[0072] Barrel **22** includes an axial bore **37** extending longitudinally and axially from a rear breech end **38** to a front muzzle end **39** of the barrel. The bore **37**, which may be rifled, defines a projectile passageway through which a bullet or slug is discharged from the firearm. The centerline of bore

37 is coaxial with and defines the longitudinal axis **LA** of the firearm. The rear breech end **38** of the barrel **22** defines a rearwardly open diametrically enlarged chamber **36** configured for holding a cartridge. Chamber **36** communicates with barrel bore **37**.

[0073] Receiver **21** defines an axially elongated internal cavity **40** which slideably carries and supports the bolt **50** (see, e.g., FIG. 3). Cavity **40** may be upwardly open for inserting the bolt therein over the rear portion of barrel **22** which may project into the cavity by greater than 25% of the length of the cavity as shown. Barrel may extend 50% or more into the length of cavity **40** in some embodiments, and may occupy a majority of the length of the cavity in one embodiment as shown.

[0074] Cavity **40** of receiver **21** extends along the longitudinal axis **LA** between the receiver open front end **18** in communication with the barrel chamber **36** for loading cartridges therein and a closed rear end **19** defined by vertical rear end wall **43**. The cavity has a sufficient length to permit the bolt **50** to cycle rearward far enough to extract and eject a cartridge from magazine **33**, and load a fresh cartridge therefrom into the barrel. Barrel **22** is coupled to the front end **18** of the receiver via any suitable method used in the art. In one embodiment, a threaded coupling may be used in which barrel **22** includes threads **18a** which rotatably engage a threaded socket **18b** formed on the receiver front end. The threaded portion of the barrel is spaced forward from its rear breech end **38** by a distance (see, e.g., FIG. 3). Accordingly, the rear breech end of the barrel is not threaded. The receiver **221** is detachably coupled to and supported by the frame **82** by any suitable means including pins and/or fasteners, interlocking tabs, etc.

[0075] For convenience of assembly and maintenance/repair, the firing mechanism **23** in one embodiment as shown may be housed and mounted in a separate fire control insert **23-1** (see, e.g., FIGS. 12 and 14-23). Fire control insert **23-1** is detachably and removably mounted to the lower portion of the mid-stock **82** such as to the pistol grip frame **82-1** in the illustrated embodiment via any suitable method used in the art such as for example without limitation via mounting pins, threaded fasteners, interlocking surfaces or protrusions, combinations thereof, or other mechanical fastening means. Other suitable type fastening means however may be used. Grip frame **82-1** defines an upwardly open longitudinal receptacle **82-2** which receives the fire control insert and certain components of the fire control system therein (see, e.g., FIG. 12). The fire control insert **23-1** includes an internal cavity **23-2** for mounting various fire control components further described herein.

[0076] In one embodiment, firearm **20** may include a rotatable manual safety mechanism **41** is transversely movably mounted in fire control insert **23-1** and configured to interact with a component of the firing mechanism for rendering the firing mechanism in either a disabled (safe) or enabled (fire) condition. Safety **41** may be an ambidextrous safety operably from either lateral side of the firearm. Such a safety may include right safety operating button **41-2** operably coupled to a left safety operating button **41-1** via a transversely mounted shaft **41-3** extending from side-to-side through the fire control insert.

[0077] Fire control insert **23-1** may be formed of any suitable metallic and/or reinforced polymeric material in some embodiments. The fire control insert has an axially elongated body defining a front end **70**, rear end **71**, right lateral side **72**, and left lateral side **73**. The fire control insert body extends along the longitudinal axis **LA** and may include features to key the insert into the pistol grip frame **82-1** for proper alignment therewith to couple these components together as described elsewhere herein.

[0078] In addition to trigger **24**, the firing mechanism **23** may include the following additional components mounted to the fire control insert **23-1**: a pivotable and cockable spring-biased hammer **25**; spring-biased pivotable sear **26** which is configured and operable to hold the hammer in a rear cocked position, and trigger bar **27** operably coupled between and to the trigger and sear for actuating the sear via a trigger pull to release the hammer from the cocked position for firing the firearm. Trigger pivot **24-1** mounts trigger **24** to the fire control insert in a pivotable manner in one

embodiment. Trigger spring **24-1** may be a torsion spring with a coil portion mounted about the trigger pin to bias the trigger **24** forward to a ready-to-fire position, and a rearward extending leg which acts on the trigger bar **27** to bias it upwards for engagement of a sear operating protrusion **27-1** at the rear end of the trigger bar with the sear.

[0079] Hammer **25** is biased forward towards a front firing position by hammer spring **31**. Sear **26** is biased towards engagement with the hammer via sear spring **26-1**. Pulling the trigger **24** rearward shifts the trigger bar **27** forward which in turn rotates the sear **26** via sear operating protrusion **27-1** on the trigger bar. This disengages a hook or ledge **28** formed on the front of the sear from sear notch **29** on the hammer **25**, thereby releasing spring-biased hammer **25** forward which strikes the rear end of firing pin **30** slidably carried by the bolt **50**. This drives the firing pin forward to strike a chambered cartridge held in the chamber **36** of the barrel **22** for discharging the firearm **20**.

[0080] Referring generally to FIGS. **3** and **8-13**, bolt **50** in one embodiment may comprise an axially elongated block-shape metal body of substantially rectilinear (e.g. rectangular cuboid) shape. The term “substantially” connotes that the overall bolt body configuration may be considered to approximate a rectangular cuboid in contrast to cylindrical firearm bolts, and excludes protruding or recessed features appended on the bolt body. Bolt **50** may have a monolithic unitary one-piece structure, or alternatively may be a two-piece construction as illustrated herein comprising a front bolt piece **50-1** and rear bolt piece **50-2** detachably coupled together to collectively form a bolt assembly. Each bolt piece may have a monolithic unitary structure in which features disclosed herein (including the keyed mutually configured male and female coupling features) are formed by any suitable mechanical fabrication means used in the art such as casting, metal injection molding, milling, boring, drilling, machining, etc. and combinations thereof. This is distinguishable from rotating bolt designs of some auto-loading rifles such as gas-operated ARs which may comprise a non-rotatable bolt carrier and a separate discrete rotatable bolt comprising the radial bolt lugs for forming a locked breech. These type bolts are not used or compatible with firearms having a blowback operating system such as disclosed herein which do not form locked breeches that no parts of the bolt are rotatable as in the present two-piece keyed bolt design.

[0081] It bears noting that reference to “bolt **50**” herein should also be considered and construed to also include reference to the entire two-piece bolt assembly as applicable; the singular term “bolt” being interchangeably used for brevity of description.

[0082] Significantly, the two-piece bolt construction advantageously provides a modular bolt system which allows the mass of the bolt assembly to be altered via swapping out only one the bolt pieces with another having a greater or less mass, which affects the recoil of the bolt mass rearward when the firearm is fired based on the loading type (powder charge) for a given caliber. The bolt therefore is customizable to ensure proper cycling of the action (i.e. fully open and reclose breech).

[0083] In one embodiment, the front bolt piece **50-1** of the bolt assembly **50** may be the piece whose mass can be varied. Accordingly, a plurality of front bolt pieces **50-1** each with similar bolt body configuration and a common coupling feature or interface but different masses may be provided which are all configured to be compatible for interchangeable use and coupling with a single rear bolt piece **50-2** having a coupling feature to which the front bolt pieces can be detachably coupled. This advantageously provides fabrication and replacement economies for the firearm manufacturer and end user thereby allowing a single firearm to accommodate a range of propellant powder charges/loads for a particular caliber of ammunition simply by swapping front bolt pieces.

[0084] Various interchangeable front bolt pieces **50-1** for bolt **50** therefore allows for configuring a given caliber within the allowable range of ammunition load offerings on the market. This also significantly simplifies the amount of alterations to the bolt system. Barrels and various other features may also need changing. It further bears noting that the rear bolt piece defines the breech face which is still often caliber specific. Accordingly, changing calibers of ammunition may require

a different rear bolt piece to match the breech face to the caliber selected.

[0085] As further explanation of the foregoing concept, blowback operating systems of firearms cycle the action by first moving the bolt first rearward to open the breech after firing to extract and eject an ammunition cartridge from the barrel cartridge chamber. The recoil spring then automatically returns the bolt forward to strip a fresh cartridge out of the magazine and chamber the cartridge as the breech is reclosed and readied for the next shot.

[0086] The rearward acting recoil forces generated by firing the firearm must be sufficient to move the bolt rearward and properly time opening of the breech by taking into account the total mass of the bolt and contribution of the forward acting spring force of recoil spring **54** which is compressed when the breech is opened. The recoil spring biases the bolt forward towards the closed breech position in opposition to the recoil forces. Accordingly, it is crucial to match the mass of the bolt to the propellant powder charge or load of the caliber of cartridge for which the firearm is chambered to ensure proper cycling of the action. If the bolt mass is too low for the cartridge, the bolt may cycle rearward too quickly which can damage the bolt or return the bolt prematurely forward to reclose the breech before a new cartridge can be full uploaded from the magazine thereby jamming the action. Conversely, if the bolt mass is too great the bolt may not cycle rearward sufficiently to fully open the breech, or too slowly which adversely affects the timing of the action. It is therefore clear that the timing of the action affected by the bolt mass is critical to proper operation of the firearm.

[0087] Accordingly, if the manufacturer offers a same firearm configured to fire a particular caliber of ammunition but with different powder charges/loads to meet consumer needs and preferences, individual entire one-piece monolithic bolts would have to be manufactured in a variety of masses. This becomes an expensive proposition for the firearm manufacturer or end user seeking to change ammunition calibers since the blowback bolts are machined parts.

[0088] The present invention provides a modular two-piece bolt system configured for blowback operation which advantageously allows the same rear bolt piece **50-2** which includes the breech face to be used for a certain caliber of ammunition but which may be offered commercially in variety of different propellant powder charges/loads. For example, 10 mm caliber cartridges have muzzle energy which drastically varies based on the particular ammunition powder charge/load selected; all within the 10 mm family. Another use case is the use of suppressors. These increase backpressure, transferring more energy to the bolt. A heavier front bolt with greater mass is desirable for use with a suppressor. Accordingly, even though the caliber and powder charge/load of the ammunition might not be changed, adding a suppressor changes the mass requirements for the front bolt and entire bolt assembly.

[0089] In both foregoing scenarios, only the front bolt piece **50-1** of the modular bolt assembly needs to be selected for use to adjust and match the collective total mass of the entire 2-piece bolt assembly to the energy (e.g., propellant powder load) generated by a particular powder charge/load of the cartridge to be fired with the firearm. In the present design, the mass of the rear bolt piece is therefore a fixed parameter. A plurality of different mass front bolt pieces may be manufactured from which the manufacturer can select and couple the appropriate one to the rear bolt piece which is compatible for firing with the powder charge/load of the particular cartridge desired.

Furthermore, the rear bolt piece may be more expensive to fabricate of the two bolt pieces since the rear bolt piece **50-2** of the bolt assembly disclosed herein contains the breech face and flanged coupling protrusion. The modular two-piece bolt system therefore provides both fabrication/supply economies and manufacturing efficiencies.

[0090] The modular bolt system is designed so that in one application an end user who initially owns a firearm with a first barrel chambered for a first cartridge of a given caliber with a powder charge/load and corresponding mass-matched 2-piece bolt **50** disclosed herein may change the cartridge to be fired of the same caliber. Accordingly, a second barrel chambered for a second cartridge of the same given caliber but with a different propellant powder charge/load (e.g., less or

greater than the first cartridge) may be retrofit and installed to replace the first barrel and the front piece **50-1** of the bolt assembly. The 2-piece bolt **50** may be removed from the firearm followed by uncoupling a first front bolt piece **50-1** having a first mass from the rear bolt piece **50-2**. The first front bolt piece **50-1** is then exchanged for a second front bolt piece having a different second mass than the first front bolt piece; the second front bolt piece being compatible for firing the second cartridge.

[0091] Accordingly, using the foregoing retrofit method or process, the same firearm may be re-used and retrofitted to change the cartridge to be fired within the same thereby offering significant cost savings to the user compared to having to purchase an entire bolt or even second firearm to fire the second cartridge.

[0092] It bears noting that in physics and engineering terms “mass” is not identical to “weight.” Mass is the amount of matter present in an object while its weight is a measure of the strength of the force with which gravity pulls on the object. The weight of an object (pound-force) is equal to the mass (slugs)×gravity (feet/second.sup.2). However, for practical purposes with gravity being considered constant, the terms mass and weight are often used interchangeably albeit technically different.

[0093] The front and rear bolt pieces **50-1**, **50-2** may be detachably coupled together in one embodiment as shown using a common coupling interface. A detachable keyed coupling may be provided in one embodiment formed by a male coupling feature on the front or rear bolt piece which creates a releasable interlock with a female coupling feature on the other one of the front or rear bolt piece. In one non-limiting embodiment, the male coupling feature may be a coupling protrusion **50-3** on the front of the rear bolt piece which forms the “key” and the female coupling feature may be a complementary configured coupling recess **50-4** formed on the rear of the front bolt piece which forms the “keyway.” Coupling protrusion **50-3** slideably engages the complementary configured coupling recess **50-4**. Any suitable configuration of coupling protrusion and recess may be used so long as the front bolt piece is axially interlocked to the rear bolt piece (i.e. the pieces cannot be axially separated in the longitudinal direction).

[0094] In one embodiment, coupling protrusion **50-3** may be U-shaped and coupling recess **50-4** may be a corresponding U-shaped channel which lockingly engages the protrusion when inserted. The coupling protrusion and recess may have a polygonal cross-sectional configuration in one embodiment which includes 90 degree corners. Other interlocking cross-sectional shaped may be used. The coupling protrusion **50-3** and coupling recess **50-4** in one embodiment may be formed as integral unitary parts of the front and rear bolt pieces **50-1**, **50-2** which may each have a monolithic structure. The coupling protrusion **50-3** may comprise an upwardly and laterally projecting flange which is vertically oriented as shown in one embodiment. Correspondingly, the U-shaped channel may be downwardly open such that the flanged protrusion is inserted upwards into the channel of the front bolt piece **50-1**. Alternatively, the U-shaped may be upwardly open and the coupling protrusion flange may project downwardly and laterally to each side.

[0095] When these keyed male and female coupling features of the front and rear bolt pieces **50-1**, **50-2** are mutually engaged, the two pieces are axially interlocked together to prevent separation in the longitudinal direction when the bolt assembly **50** is axially moved (i.e. cycled) in receiver **21** between the rearward open breech position and forward closed breech position in battery with the barrel. In other embodiments, the keyed coupling features may be reversed such that coupling protrusion **50-3** may instead be disposed on the rear of front bolt piece **50-1** and the coupling recess **50-4** (e.g., channel) may be disposed on the front of rear bolt piece **50-2**. Either arrangement may be used and does not limit the invention.

[0096] It bears noting that the front and rear bolt pieces physically restrained from separation when positioned in internal cavity **40** of receiver **21**. In the axial (longitudinal) direction, the two bolt pieces are keyed into one another by the interlocked coupling protrusion and recess thereby axially preventing separation. The bolt assembly once mounted in receiver **21** is also vertically restrained

from separation being disposed between the receiver above and fire control insert **23-1** below. As a result, there are no fasteners or screws required to hold the front and rear bolt sections together in the present embodiment when the bolt assembly is mounted in the firearm.

[0097] For convenience of general reference, the collective assembled bolt (bolt assembly) **50** (with front bolt piece **50-1** coupled to rear bolt piece **50-2**) includes a front end **52**, rear end **53**, top **56**, bottom **57**, and a pair of longitudinally-extending opposing right and left lateral sides **58** defined by sidewalls of the bolt body. A majority of the sides **58** may be planar in some embodiments. For the illustrated two-piece bolt construction, these foregoing general positional locations refer to the assembled bolt, not each bolt piece alone.

[0098] Front and rear bolt pieces **50-1**, **50-2** of bolt **50** each comprise a front and rear longitudinal passage **50-5**, **50-6** respectively which are coaxial with longitudinal axis LA of the firearm. The rear breech end **38** portion of barrel **22** is insertable into the passages from the open front end of the receiver **21**. Longitudinal front and rear passages **50-5**, **50-6** may each comprise an at least partially circular arcuately curved cross-sectional shape in portions to conform to the circular outer surface of the rear breech portion of the barrel inserted therein (see, e.g., FIGS. **12-13** and bolt **50A** in FIGS. **38-39**). The longitudinal passages **50-5** and **50-6** collectively form a continuous extended longitudinal passage spanning between and having portions lying in each of the front and rear bolt pieces as shown.

[0099] The front longitudinal passage **50-5** extends completely through front bolt piece **50-1** between front end **50-1A** and rear end **50-1B**. The rear longitudinal passage **50-6** extends through front end **50-2A** of rear bolt piece **50-1** and terminates rearward at a vertical breech face **50-7** disposed partway inside the longitudinal passage **50-6**, which is spaced inward from front end **50-2A** of the rear bolt piece (see, e.g., FIG. **13**). Viewed another way, breech face **50-7** is located between the front and rear ends of rear bolt piece **50-2** and spaced forward from rear end **50-2B** of rear bolt piece by a distance.

[0100] The foregoing barrel and bolt arrangement provides a bolt-over-barrel design. Extending the barrel **22** for a distance inside the receiver and bolt **50** via longitudinal passages **50-5**, **50-6** advantageously allows the forward projected length of the barrel to be short for a carbine type firearm, as previously described herein. The vertical breech face **50-7** of the rear bolt piece **50-2** engages the rear breech end **38** of the barrel **22** adjacent the cartridge chamber **36** when the breech is closed. The bolt breech face **50-7** establishes a rear terminal end of the longitudinal passage **50-5** in the rear bolt piece.

[0101] Bolt **50** (i.e. bolt assembly) is received and axially slideably movable in the internal cavity **40** of the receiver **21** along longitudinal axis LA between forward closed breech and rearward open breech positions. In the closed breech position, the bolt **50** (i.e. breech face **50-7** of rear bolt piece **50-2**) is in battery with the rear breech end **38** of barrel **22** to close the chamber **36** for firing. In the open breech position, the bolt is axially displaced rearward to allow a spent cartridge casing to be extracted from the barrel chamber and ejected, and for loading a fresh cartridge into the breech area of the receiver for chambering by the bolt when automatically returned forward to the closed breech position by the recoil spring assembly.

[0102] To facilitate sliding engagement with internal cavity **40** of receiver **21** and to support the each of the bolt pieces **50-1**, **50-2**, each of the front and rear bolt pieces may comprise a laterally spaced apart pair of downwardly-protruding linear bottom longitudinal support surfaces **59-1** (see, e.g., FIG. **13**) configured to slideably engage the flat bottom surface in the cavity **40** of the receiver. The support surfaces **59-1** may extend for at least part of the axial length of the front and rear bolt pieces. In other embodiments, the bottom surfaces of the front and/or rear bolt pieces may be a planar.

[0103] The recoil spring assembly includes recoil spring **54** and longitudinally-extending spring guide rod **55** which supports and guides the spring. The spring may be a helical compression spring in one embodiment as shown. The rear end of guide rod **55** extends rearward beyond rear bolt piece

50-2 through upwardly open longitudinal recess **59** thereof and is affixed to the rear end of receiver **21** inside receiver longitudinal cavity **40** in a fixed position. The front end of guide rod **55** is slideably received inside axial guide rod passage **50-8** of the front bolt piece **50-1** and projectible forward and rearward in the passage when the bolt moves between its forward and rearward positions. In one embodiment, recoil spring **54** may act on rear end **50-1B** of front bolt piece **50-1** alone at the rear entrance to the guide rod passage to bias the entire two-piece bolt assembly forward to the closed breech position. Cavity **40** of receiver **21** therefore has an axial length sufficient to provide the full range of motion necessary for the bolt **50** moving rearward under recoil to open the breech for extracting and ejecting a spent cartridge casing, and moving forward to the closed breech position for feeding a new cartridge into the barrel chamber **36** from the magazine **33**.

[0104] Bolt **50** may also include a manual charging handle **51** affixed to one side of the bolt to manually cycle the bolt between its forward and rearward positions by hand thereby allowing a user to manually lock the bolt in the rearward open breech position using bolt catch **100**, as further described herein. Handle **51** has a transversely elongated body and is laterally projected from one side of the firearm for grasping to cycle the action.

[0105] Firing pin **30** of the firing mechanism **23** is movably mounted in a longitudinally-extending firing pin channel **30-1** formed in the bolt body. In one embodiment, firing pin **30** is mounted in rear bolt piece **50-2**. The rear end of firing pin **30** remains exposed at the rear end **53** of the rear bolt piece (see, e.g. FIG. 3) for striking by the hammer **25** to discharge the firearm via a trigger pull. The front end of firing pin **30** is axially projectible through firing pin opening **30-2** formed through the breech face **50-7** of rear bolt piece **50-2** to strike a chambered cartridge when the breech face is in battery with the rear breech end of barrel **22**.

[0106] FIGS. **34-43** show an alternative embodiment of a bolt **50A** (bolt assembly) in which the front and rear bolt pieces **50-1**, **50-2** are slightly different in configuration. Bolts **50** and **50A** are ostensibly the same with respect to the significant technical features described herein (e.g., bolt coupling/interlock system, breech face location, longitudinal passages, recoil spring rod passage, etc.). Physical overall shape of the front bolt piece in particular is slightly different. One additional difference is that the recoil rod passage **50-8** of front bolt piece **50-1** has a closed front end and open rear end for bolt **50A** whereas this passage had open front and rear ends for prior bolt **50**. Bolt **50A** demonstrates that minor variations in the bolt shape are possible as may be needed to fit the internal geometry of the receiver cavity which holds the bolt while still retaining the key functional features and economies of the 2-piece bolt assembly for blowback operation previously described herein.

[0107] The manually actuated bolt release mechanism of the present invention and related method for operating the same will now be further described.

[0108] The bolt release mechanism comprises bolt catch **100** which cooperates and operably interfaces directly with the bolt **50** for locking and retaining the bolt in the rear open breech position, or releasing the bolt forward therefrom to close the breech. FIGS. **24-33** show details of bolt catch assembly in isolation. FIGS. **14-23** show the bolt catch assembly in association with the fire control insert **23-1** to which the bolt catch is operably mounted.

[0109] Bolt catch **100** in one embodiment is an assembly comprising bolt catch lever **101** and actuator plunger **120** which cooperates with the lever to form an ambidextrous bolt release. These components may be movably mounted to and supported by the fire control insert **23-1** of the firearm in one embodiment as shown.

[0110] Bolt catch lever **101** comprises an elongated metallic body **100-1** which lies in a vertical reference plane VP oriented parallel to the sides of the firearm (see, e.g., FIGS. **10-11**). The bolt catch lever **101** is movably disposed on one lateral side of the firearm (e.g., right lateral side **21-1** or left lateral side **21-2** of receiver **21** as shown). The bolt catch lever may be substantially axially elongated in the direction of the longitudinal axis LA of the firearm having an axial length which

extends forward of trigger **24** in the fire control insert **23-1** at front and at rear laterally adjacent to central opening **115** of fire control insert **23-1** located directly above and in communication with magazine well **32**, as further described herein. Bolt catch lever **101** is rotatable and pivotably movable about its horizontal and laterally oriented pivot axis PA upwards and downwards with toggle-like action in the vertical reference plane VP, as further described herein. The elongated extension of the bolt catch lever forward of pivot axis also provides the benefit of improving inertial balance of the lever, thereby contributing to a balanced and positively acting lever.

[0111] The bolt catch lever **101** is therefore vertically and pivotably movable about its pivot axis PA in the vertical reference plane VP relative to the bolt **50** and receiver **21** between an upward engaged position to lock and hold the bolt in its rearward open breech position, and a downward disengaged position which disengages and releases the bolt forward. Operation of the bolt catch **100** is further described elsewhere herein.

[0112] Bolt catch lever **101** generally defines a front end **102**, rear end **103**, exterior and interior sides **104**, **105** extending therebetween, top **106**, and bottom **107**. Sides **104**, **105** may be substantially flat portions of the body **100-1** (except for operating features) and parallel to each other in one embodiment as shown.

[0113] The operating features of bolt catch lever **101** generally includes a pivot pin **110** which pivotably mounts the lever to fire control insert **23-1**, bolt catch protrusion **113** configured to selectively engage a mating locking ledge **114** formed on the bolt **50** to lock the bolt rearward with an open breech until manually released, actuating handle **111** for manually actuating the lever to either manually engage or release the bolt **50**, and follower engagement protrusion **112** which is selectively engageable with the follower **33-1** of the magazine **33** as a last round bolt hold open feature after the last cartridge is dispensed to the firearm from the magazine which automatically locks the bolt in the rear open breech position. Each feature is further described below.

[0114] Pivot pin **110** extends perpendicularly to the bolt catch lever body **100-1** and transversely inwards from interior side **105** of the bolt catch lever **101** in a horizontal orientation. Pivot pin **110** in one non-limiting embodiments may be located in the middle portion of the lever body **100-1** between ends **102**, **103**, but not necessary at the midpoint therebetween as shown. Pivot pin **110** defines pivot axis PA of the lever **101** and may have a cylindrical configuration. Pivot pin **110** is rotatably received in a mating round pin mounting hole **110-1** formed in a lateral side of the fire control insert **23-1** to pivotably mount the bolt catch lever to the insert (see, e.g., FIG. **14**).

[0115] Bolt catch protrusion **113** is configured to protrude both upwards and laterally/transversely inwards from interior side **105** of the bolt catch lever **101** from the rear portion **103-1** of the lever defined between pivot pin **110** and rear end **103**. In one embodiment, as shown, bolt catch protrusion **113** may be disposed proximate to rear end **103** of the lever to maximize its vertical range of motion about the lever pivot axis PA. The bolt catch protrusion is configured to operably engage locking ledge **114** formed on the bolt **50** to lock the bolt in its rear open breech position when bolt catch lever **101** is either manually actuated by the user or automatically actuated via operation of the follower engagement protrusion **112** operably interfaced with the magazine as a last round hold open provision.

[0116] In one embodiment shown in FIGS. **8-9** and **13**, locking ledge **114** may be formed on the underside of one of the lateral sides **58** of the bolt body. In the present embodiment, locking ledge **114** is formed on the bottom of the left lateral side of bolt **50**. Locking ledge **114** defines a forward facing locking surface **114-1** which engages the bolt catch protrusion **113** when the bolt catch lever **101** is actuated and in its upward engaged position. Locking surface **114-1** may be formed at the rear end of a downwardly open longitudinal recess **114-2** formed forward of the locking surface on the bolt. In one embodiment, longitudinal recess **114-2** may be formed on the underside of the left lateral side of the bolt **50** and slideably receives the bolt catch protrusion **113** when the bolt catch protrusion is moved upwards with the rear end of the bolt catch lever as it pivots to its upward engaged position and the bolt is attempting to be returned forward by the recoil spring from the

bolt's rear open breech position. Bolt catch protrusion **113** slides rearward in the recess until it meets and engages the locking ledge **114** and locking surface **114-1**, which thereby retains the bolt in the rearward open breech position.

[0117] In one embodiment, the locking ledge **114** and longitudinal recess **114-2** may be formed on the rear bolt piece **50-2** for the two-piece bolt **50** construction. The locking ledge is positioned on the bolt relative to the bolt catch protrusion **113** of the bolt catch lever **101** to establish and retain the open breech until the bolt is manually released forward by the user via actuating handle **111** or actuator plunger **120**.

[0118] Actuating handle **111** protrudes laterally/transversely outwards from exterior side **104** of the bolt catch lever **101** from the rear portion **103-1** of the lever defined between pivot pin **110** and rear end **103**. The actuating handle may be disposed proximate to rear end **103** of the bolt catch lever **101**, and in one embodiment may define the rear end. Actuating handle **111** may be disposed adjacent to and on the opposite side of the bolt catch protrusion **113** on the lever to maximize leverage about the pivot axis PA and range of vertical motion of the bolt catch protrusion to positively engage the bolt when actuated. The actuating handle may have any suitable configuration to enhance grasping by the user to manually actuate the bolt catch lever **101** for either engaging and locking the bolt **50** in its rearward open breech position, or disengaging and releasing the bolt forward. Handle **111** is pushed upward to engage the bolt, and pushed downwards by the user to disengage the bolt.

[0119] Follower engagement protrusion **112** of bolt catch lever **101**, which is configured and arranged to be automatically engaged by follower **33-1** of the magazine **33** when empty, protrudes perpendicularly and laterally/transversely inwards in a horizontal direction from interior side **104** of the bolt catch lever **101**. Follower engagement protrusion **112** may be disposed on the rear portion **103-1** of the lever defined between pivot pin **110** and rear end **103**. The follower engagement protrusion may be located forward of actuating handle **111** on bolt catch lever **101** in one embodiment. Follower engagement protrusion **112** may have a transversely elongated and can have a generally flattened body in one embodiment.

[0120] With additional reference to FIGS. **14-23** as applicable, follower engagement protrusion **112** extends inwards into a central opening **115** of fire control insert **23-1** which is located directly above and communicates with magazine well **32** of the pistol grip frame **82-1**. The follower engagement protrusion may be located proximate to the front of central opening **115** and magazine well **32** for engagement by the magazine follower **33-1** after the last cartridges has been dispensed from the magazine. The follower engagement protrusion therefore extends inwards into the central opening **115** by a sufficient distance to positively engage the magazine follower **33-1** when no rounds remain in the magazine.

[0121] After the last chambered round (cartridge) is fired from the magazine **33**, the magazine follower **33-1** forced upwards by magazine spring **33-2** will protrude up into the breech area of receiver **21** from the top of the magazine to engage and force the follower engagement protrusion **112** of bolt catch lever **101** upwards. This pivots the rear end **103** of the bolt catch lever in turn about its pivot axis PA (defined by pivot pin **110**) which moves from the downward disengaged position to the upward engaged position. This motion in turn raises the bolt catch protrusion **113** upward to engage the locking ledge **114** of the bolt as it attempts to return forward under the biasing action of recoil spring **54**. The breech is therefore locked in the open position via automatic operation of the magazine follower **33-1** and follower engagement protrusion **112**. To release the bolt, the user manually depresses actuating handle **111** downward as previously described herein, or alternatively uses actuator plunger **120**.

[0122] Catch spring **116** is configured and operable to bias bolt catch lever **101** towards its downward disengaged position to prevent interference with cycling of bolt **50** rearward and forward under normal operation of the firearm when firing until the bolt catch lever is activated. The catch spring **116** may be a torsion spring in one embodiment which is mounted about pivot pin

110. Spring **116** may comprise a rear partial coil section **116-1** which engages and partially wraps around the pivot pin and a forward extending leg **116-2** which acts downward on magazine-operated follower engagement protrusion **112** to bias the bolt catch lever downwards (see, e.g., FIG. **28**). Other types of springs and arrangements may be used to provide the desired biasing operation.

[0123] Actuator plunger **120** provides manual ambidextrous operation of the bolt-catch release mechanism from the opposite side of the firearm on which the bolt catch lever **101** is disposed. Specially, the actuator plunger when depressed is therefore configured and operable to engage and actuate the bolt catch lever, thereby causing the lever to disengage and release the bolt **50** when locked rearward by the bolt catch without use of the actuating handle **111** on the bolt catch lever.

[0124] Referring generally to FIGS. **24-33**, the manually depressible spring-biased actuator plunger **120** may be transversely mounted in the firearm relative to the longitudinal axis LA (e.g., perpendicularly) and extends through the firearm from one lateral side **21-1** (e.g., right side) to the opposite lateral side **21-2**. In one embodiment, plunger **120** may be transversely mounted to and slideably received through a laterally open transverse mounting passage **121** formed through the front portion of fire control insert **23-1** from side to side (see also FIGS. **14-23**).

[0125] As shown in FIG. **32**, actuator plunger **120** is oriented perpendicularly to the bolt catch lever **101**. The elongated body of bolt catch lever **101** defines a lever axis AX1 and plunger **120** defines a plunger axis AX2 which is oriented perpendicularly to the lever axis. Plunger **120** operably interacts with the front portion **103-2** of bolt catch lever **101**, and in a preferred but non-limiting embodiment shown engages the lever proximate to the front end **102** thereof, as further described herein.

[0126] Actuator plunger **120** is slideably moveable transverse to the longitudinal axis LA of the firearm between (1) an actuated position engaged with the bolt catch lever **101** (see, e.g., FIG. **11**), and (2) an unactuated position (see, e.g., FIG. **10**) disengaged from the bolt catch lever to avoid interference with operation of the lever when actuated manually via actuating handle **111** or magazine-operated follower engagement protrusion **112**. In the “engaged” position, the plunger **120** actively displace and pivots the bolt catch lever **101** about its pivot axis PA to activate the bolt stop function. In the “disengaged” position, the plunger does not engage altogether or does not sufficiently engage the bolt catch lever to the degree necessary to impart motion to and actuation of the bolt catch lever to release the bolt (although some minor contact between these parts may exist). The actuator plunger **120** converts sliding linear motion of the plunger into rotational/pivoting motion of the bolt catch lever **101** about its pivot axis PA.

[0127] In one embodiment, actuator plunger **120** comprises a generally cylindrical body **120-1** comprising an externally exposed operating end **122** configured for operating the plunger by the user, and an opposite working end **123** configured to selectively engage and rotate the bolt catch lever when the plunger is actuated. Operating end **122** may be diametrically enlarged in one embodiment defining an enlarged circular head **124** which protrudes outwards to facilitate ease of operation by the user for depressing the plunger towards the bolt catch lever **101**. In other embodiments, the operating end may not be diametrically enlarged.

[0128] Plunger spring **130** biases the plunger **120** towards the unactuated position “operably” disengaged from the bolt catch lever **101** as explained above. Spring **130** may be a helical compression spring in one embodiment which is disposed inside the transverse mounting passage **121** of fire control insert **23-1** and coiled around the plunger body (see, e.g., FIGS. **10-11**). In one embodiment, plunger **120** includes diametrically enlarged travel stop flange **125** which is selectively engageable with a pair of opposing travel stop surfaces **131** and **132** formed inside mounting passage **121**. The flange **125** may be slideably disposed inside a corresponding diametrically enlarged chamber **133** of the mounting passage which defines stop surfaces **131**, **132** at opposite sides thereof as shown. Travel stop flange **125** abuttingly engages travel stop surface **131** when the plunger is not actuated (i.e. depressed inwards by the user). When the user depresses

plunger **120**, the stop flange engages the opposite travel stop surface **132**. The travel stop flange and stop surfaces limit the lateral/transverse range of motion of the plunger **120**.

[0129] The working end **123** of actuator plunger **120** is configured to operably interact with and impart motion to the bolt catch lever **101** to disengage and release bolt in the manner previously described herein when the plunger is depressed inwards by the user towards the lever. In one embodiment, the user applies a non-rotating inward linear force F as shown in FIG. **11** to the operating end **122** (e.g., enlarged circular head **124**) of the plunger. This slides the plunger transversely and perpendicularly towards bolt catch lever **101**. Mutually engaged sliding surfaces of the plunger and bolt catch lever respectively operably cooperate to actuate and rotate the bolt catch lever. It bears noting that there is no abutting engagement between the plunger and bolt catch lever when the plunger is depressed and actuated which would force the lever laterally outwards from the firearm. Instead, a lifting motion is applied to the front end **102** of bolt catch lever **101** by the actuator plunger **120** via a pair of mating inclined surfaces **126**, **131** which slide relative to each other, as described below.

[0130] With continuing reference to FIGS. **24-33** and also FIGS. **10-11**, working end **123** of actuator plunger **120** includes an inclined camming surface **126** which slideably engages a corresponding inclined camming ramp **131** formed on the bolt catch lever **101** to pivot the bolt catch lever from the engaged position to the disengaged position when the actuator plunger is depressed and actuated. In one embodiment, the plunger working end **123** may have a configuration forming a conically shaped tip which defines a circumferential continuum of inclined camming surfaces **126** on the working end having the same angle of inclination or slope. With this conical tip design, any slight rotation of the cylindrical-bodied plunger about its plunger axis $AX2$ which might possibly occur when the user pushes the plunger inwards will advantageously ensure that inclined camming surface **126** is always properly oriented with respect to the inclined camming ramp **131** on bolt catch lever **101** to ensure reliable operation of the bolt-catch release mechanism. This design also obviates any need for anti-rotation features to be incorporated into the plunger mechanism to ensure proper alignment and orientation of the plunger camming surface **126** with the bolt catch lever camming ramp **131**.

[0131] In other possible embodiments contemplated, however, actuator plunger **120** may have a non-cylindrically shaped body and/or may include anti-rotation features if provided with a single inclined camming surface on the plunger working end **123** of fixed orientation rather than a conical tip.

[0132] It bears noting that in yet other possible embodiments, only one inclined surface such as camming surface **126** on plunger **120** or the camming ramp **131** bolt catch lever **102**.

[0133] The inclined camming ramp **131** on bolt catch lever **101** may be disposed on a front portion **103-2** of the bolt catch lever forward of the pivot axis PA . In one embodiment, camming ramp **131** may be formed on the underside/bottom of the front end **102** of the bolt catch lever near the tip. This advantageously maximizes the length of the lever arm between the camming ramp and pivot axis PA in the middle section of the lever (formed by pivot pin **110** which acts as a fulcrum) so that raising the front end of the bolt catch lever via operation of actuator plunger **120** concomitantly lowers the bolt catch protrusion **113** formed on rear end **103** of the lever with sufficient force to ensure positive disengagement with and release of the spring-biased bolt **50** forward.

[0134] In one embodiment, with reference to FIGS. **10-11** and **31**, the inclined camming ramp **131** of bolt catch lever **101** may slope in the lateral direction of the plunger axis $AX1$ and a transverse to both vertical reference plane VP of the bolt catch lever and longitudinal axis LA of the firearm. The low point on the ramp is therefore toward and adjacent the exterior side **104** of bolt catch lever **101**, while the high point is adjacent the interior side **105** of the bolt catch lever. Inclined camming surface **126** of the plunger **120** is sloped in a similar direction and orientation with the low point on the surface being adjacent to the end of the pointed conical tip of the working end **123** of the plunger. The inclined camming surface **126** and ramp **131** have complementary angles of

inclination so that a flat-to-flat sliding interface is formed therebetween in operation of the plunger to raise the front end **102** of bolt catch lever **101** and rotate the rear end **103** of the lever downwards about pivot pin **110**. Accordingly, actuator plunger is operable to slideably engage the bolt catch lever when the user actuates the plunger.

[0135] In other possible alternative embodiments, the inclined camming ramp **131** of bolt catch lever **101** may alternatively be sloped in the axial/longitudinal direction on the underside of the lever front end **102** (i.e. front to rear of the lever) to engage the camming surface **126** formed on the pointed conical tip of the plunger working end **123**. Wedging action imparted to the bolt catch lever by actuator plunger **120** which operates to rotate the lever between its engaged and disengaged positions as previously described herein would function in the same way.

[0136] Bolt catch lever **101** and actuator plunger **120** preferably may each be formed of a suitable metallic material such as steel, aluminum, titanium, or other. Bolt catch lever **101** in one non-limiting embodiment may have a one-piece monolithic unitary structure in one embodiment wherein each of the foregoing operating features **110-113** previously described herein are formed as integral parts thereof. Plunger **120** may have a similar one-piece monolithic unitary structure.

[0137] A process or method for operating bolt catch **100** of a firearm **20** will now be briefly described with reference to the figures. The firearm may begin in a ready-to-fire position with a closed breech shown in FIG. **3**. Bolt **50** is in battery with the rear breech end of the barrel **22** under the forward biasing action of recoil spring **54**. The bolt catch lever **101** of the bolt catch in the normal downward and lower disengaged position from bolt **50** under the biasing action of catch spring **116**. Accordingly, bolt catch protrusion **113** is not projected upward to engage the bolt which would might interfere with normal cycling of the action (e.g., bolt **50**) forward and rearward when discharging the firearm.

[0138] Operation of the bolt catch **100** starts with first moving bolt **50** from the forward closed breech position in battery with the barrel to the rearward open breech position (see, e.g., FIG. **8A**). This can be performed either manually by the user pulling rearward on charging handle **51** to retract the bolt, or automatically via discharging with firearm which automatically moves the bolt rearward under recoil or other means such as a gas system.

[0139] With the bolt in the rearward open breech position, the method continues with engaging the bolt catch lever **101** with the bolt **50** to hold the bolt rearward. Specially, this involves rotating bolt catch protrusion **113** on lever **101** upwards about the lever pivot axis PA in a first direction to engage locking ledge **114** on the bolt. This can performed manually by the user pushing actuating handle **111** upwards to engage bolt catch protrusion **113** with the bolt, or automatically via operation of the last round bolt “hold open” feature previously described herein (i.e. magazine follower **33-1** engaging and forcing the follower engagement protrusion **112** of bolt catch lever **101** upwards to engage the bolt catch protrusion with the bolt). Either operating scenario results in the bolt **50** being locked and held rearward with an open breech.

[0140] To release the bolt **50** from the side of the firearm opposite the bolt catch lever, the method continues with the user pushing and sliding the actuator plunger **120** in a laterally inwards transverse direction towards to the bolt catch lever **101** from an unactuated position to an actuated position. This results in slideably engaging inclined camming surface **126** on the actuator plunger **120** with an inclined camming ramp **131** on the bolt catch lever, thereby rotating the bolt catch lever about the pivot axis PA in a second direction which disengages and releases the bolt from the open breech position. Specifically, “wedging” action applied to the underside of the front end **102** of the bolt catch lever via mutual sliding engagement between the inclined camming surface and ramp raises the lever front end and correspondingly forces the opposite rear end **103** downwards about pivot axis PA to disengage the bolt catch protrusion **113** from locking ledge **114** on bolt **50**, thereby returning the bolt forward to reclose the breech via the biasing action of recoil spring **54** previously described herein.

[0141] Although bolt catch lever **101** is illustrated and described herein as being located on the left

side of the firearm (i.e. left lateral side **21-2** of receiver **21**) and the operating end **122** (e.g., enlarged head **124**) of actuator plunger **120** is on the right lateral side **21-1**, these locations may be reversed in other embodiments provided the locking ledge **144** of bolt **50** is provided on the opposite side from that illustrated to allow the bolt catch to lock the bolt rearward. Accordingly, the invention is not limited to either arrangement which will function the same as described herein. [0142] While the foregoing description and drawings represent preferred or exemplary embodiments of the present invention, it will be understood that various additions, modifications and substitutions may be made therein without departing from the spirit and scope and range of equivalents of the accompanying claims. In particular, it will be clear to those skilled in the art that the present invention may be embodied in other forms, structures, arrangements, proportions, sizes, and with other elements, materials, and components, without departing from the spirit or essential characteristics thereof. In addition, numerous variations in the methods/processes as applicable described herein may be made without departing from the spirit of the invention. One skilled in the art will further appreciate that the invention may be used with many modifications of structure, arrangement, proportions, sizes, materials, and components and otherwise, used in the practice of the invention, which are particularly adapted to specific environments and operative requirements without departing from the principles of the present invention. The presently disclosed embodiments are therefore to be considered in all respects as illustrative and not restrictive, the scope of the invention being defined by the appended claims and equivalents thereof, and not limited to the foregoing description or embodiments. Rather, the appended claims should be construed broadly, to include other variants and embodiments of the invention, which may be made by those skilled in the art without departing from the scope and range of equivalents of the invention.

Claims

1. A firearm with bolt assembly for a blowback operating system comprising: a longitudinal axis; a receiver defining an axially elongated internal cavity; a barrel supported by the receiver, the barrel defining an axial bore extending longitudinally from a rear breech end to a front muzzle end of the barrel; a bolt assembly slideably disposed in the cavity of the receiver, the bolt assembly axially movable between a forward closed breech position in battery with the rear breech end of the barrel and a rearward open breech position; a recoil spring biasing the bolt forward towards the closed breech position; the bolt assembly comprising a front bolt piece detachably coupled to a rear bolt piece; the front and rear bolt pieces each comprising a longitudinal passage coaxial with the longitudinal axis of the firearm, the longitudinal passage of the front bolt piece in communication with the longitudinal passage of the rear bolt piece to collectively form a continuous extended longitudinal passage; wherein a rear breech end of the barrel is disposed inside the extended longitudinal passage.
2. The firearm according to claim 1, wherein the longitudinal passages of the front and rear bolt pieces are formed internally in the front and rear bolt pieces.
3. The firearm according to claim 2, wherein the rear bolt piece comprises a breech face disposed in the longitudinal passage of the rear bolt piece, the breech face being movable into and out of battery with the rear breech end of the barrel when the bolt assembly is moved between the open breech position and the closed breech position.
4. The firearm according to claim 3, wherein the rear bolt piece comprises a firing pin which is projectible through a firing pin opening in the breech face.
5. The firearm according to claim 3, wherein the bolt assembly is configured so as to be non-rotatable in the receiver.
6. The firearm according to claim 3, wherein the longitudinal passage of the rear bolt piece is downwardly open.

7. The firearm according to claim 3, wherein each of the front and rear bolt pieces has a monolithic construction.
8. The firearm according to claim 2, wherein the recoil spring is a helical compression spring mounted on a guide rod which is slideably received inside an axial guide rod passage of the front bolt piece.
9. The firearm according to claim 8, wherein the guide rod is projectible forward and rearward in the guide rod passage when the bolt assembly moves between the open breech position and the closed breech position.
10. The firearm according to claim 9, wherein the guide rod is fixedly attached to the receiver and remains stationary when the bolt assembly moves between the open breech position and the closed breech position.
11. The firearm according to claim 9, wherein the recoil spring acts on a rear end of the front bolt piece alone at a rear entrance to the guide rod passage to bias the entire bolt assembly forward to the forward closed breech position.
12. The firearm according to claim 3, wherein the front and rear bolt pieces are keyed into one another by a coupling protrusion disposed on the front or rear bolt piece interlocked with a mating coupling recess disposed on the other of the front or rear bolt piece.
13. The firearm according to claim 12, wherein the front and rear bolt pieces are axially interlocked together to prevent separation in the longitudinal direction when the bolt assembly is moved between the rearward open breech position and forward closed breech position.
14. The firearm according to claim 12, wherein the coupling protrusion is disposed on a front of the rear bolt piece and the coupling recess is disposed on a rear of the front bolt piece.
15. The firearm according to claim 13, wherein the coupling protrusion is U-shaped and slideably engages the coupling recess which is a complementary configured U-shaped channel.
16. The firearm according to claim 15, wherein the U-shaped coupling channel is downwardly open such that the U-shaped coupling protrusion slides upwards into the coupling channel when the front bolt piece is coupled to the rear bolt piece.
17. The firearm according to claim 15, wherein the coupling protrusion comprises a flange which is integrally formed with the rear bolt piece as a unitary structural part thereof.
18. The firearm according to claim 13, wherein the front bolt piece is coupled to the rear bolt piece without use of fasteners.
19. The firearm according to claim 5, wherein each of the front and rear bolt pieces comprise a pair of linear bottom longitudinal support surfaces configured to slideably engage a bottom surface in the internal cavity of the receiver.
20. The firearm according to claim 19, wherein each of the front and rear bolt pieces has a substantially rectangular cuboid configuration.
21. The firearm according to claim 5, wherein the breech is not mechanically lockable with the bolt assembly when the bolt assembly is in the forward closed breech position in battery with the rear breech end of the barrel.
22. A bolt assembly for a firearm with blowback operating system comprising: a longitudinal axis; an axially elongated front bolt piece internally comprising a first longitudinal passage coaxial with the longitudinal axis, the first longitudinal passage being configured to receive a barrel of the firearm; an axially elongated rear bolt piece internally comprising a second longitudinal passage coaxial with the longitudinal axis, the second longitudinal passage being configured to receive the barrel of the firearm; the second longitudinal passage being in communication with the first longitudinal passage of the front bolt piece to collectively form a continuous extended longitudinal passage; the front bolt piece being detachably coupled to the rear bolt piece and interlocked together to prevent separation in a longitudinal direction; a breech face formed in the second longitudinal passage of the rear bolt piece.
23. The bolt assembly according to claim 22, wherein the front and rear bolt pieces are keyed into

one another by a coupling protrusion disposed on the front or rear bolt piece interlocked with a mating complementary configured coupling recess disposed on the other of the front or rear bolt piece.

24. The bolt assembly according to claim 23, wherein the coupling protrusion is disposed on a front of the rear bolt piece and the coupling recess is disposed on a rear of the front bolt piece.

25. The bolt assembly according to claim 24, wherein the coupling protrusion is integrally formed as a unitary structural part of the rear bolt piece, and the coupling recess is integrally formed as a unitary structural part of the front bolt piece.

26. The bolt assembly according to claim 23, wherein the coupling protrusion is U-shaped and slideably engages the coupling recess which is a complementary configured U-shaped channel.

27. The bolt assembly according to claim 26, wherein the U-shaped coupling channel is downwardly open such that the U-shaped coupling protrusion slides upwards into the coupling channel when the front bolt piece is coupled to the rear bolt piece.

28. The bolt assembly according to claim 26, wherein the coupling protrusion comprises a flange which is integrally formed with the rear bolt piece as a unitary structural part thereof.

29. The bolt assembly according to claim 22, wherein the front bolt piece is coupled to the rear bolt piece without use of fasteners.

30. The bolt assembly according to claim 29, wherein each of the front and rear bolt pieces comprise a pair of linear bottom longitudinal support surfaces configured to slideably engage a bottom surface in the internal cavity of the receiver.

31. The bolt assembly according to claim 22, wherein each of the front and rear bolt pieces has a substantially rectangular cuboid configuration.

32. The bolt assembly according to claim 22, wherein the first longitudinal passage extends completely through the front bolt piece, and the second longitudinal passage extends partially rearward from a front end of the rear bolt piece to a point terminated by a vertical breech face inside the second longitudinal passage.

33. The bolt assembly according to claim 32, wherein the breech face is configured to engage a rear breech end of a barrel defining a chamber for an ammunition cartridge.

34. The bolt assembly according to claim 22, wherein the front bolt piece comprises a longitudinally-extending axial guide rod passage configured to receive a recoil spring rod therein.

35. The bolt assembly according to claim 34, wherein the guide rod passage comprises a closed front end and an open rear end.

36. A modular bolt assembly system for a firearm with blowback operating system comprising: a longitudinal axis; a first front bolt piece having a first mass and internally comprising a first longitudinal passage coaxial with the longitudinal axis, the first longitudinal passage being configured to receive a rear portion of a first firearm barrel chambered for a first cartridge; a second front bolt piece having a second mass different than the first mass, the second front bolt piece internally comprising a second longitudinal passage coaxial with the longitudinal axis, the second longitudinal passage being configured to receive a rear portion of a second firearm barrel chambered for a second cartridge having a propellant powder charge with energy different than the first cartridge; a rear bolt piece internally comprising a third longitudinal passage coaxial with the longitudinal axis, the third longitudinal passage comprising a breech face and being configured to receive the rear portions of the first or second firearm barrel; wherein the first and second front bolt pieces each have a common first coupling interface configured to mate with a corresponding complementary configured second coupling interface formed on the rear bolt piece; wherein the rear bolt piece is detachably coupleable to the first or second front bolt piece to form a mechanical interlock which prevents separation thereof in a direction along the longitudinal axis.

37. The system according to claim 36, wherein the third longitudinal passage of the rear bolt piece is in communication with the first longitudinal passage or the second longitudinal passage depending on whether the first front bolt piece or the second front bolt piece is coupled to the rear

bolt piece.

38. The system according to claim 36, wherein the first and second front bolt pieces each have a similar configuration.

39. The system according to claim 38, wherein the common first coupling interface of the first and second front bolt pieces comprises: (1) a male key which is insertable into a female keyway of the rear bolt piece; or (2) a female keyway which receives a male key of the rear bolt piece.

40. The system according to claim 39, wherein the male key comprises a U-shaped coupling protrusion and the female keyway comprises a complementary configured U-shaped coupling channel into which the coupling protrusion is slideably insertable.

41. The system according to claim 40, wherein the coupling protrusion is disposed on a front of the rear bolt piece and the coupling channel is disposed on a rear of the first and second front bolt pieces.

42. The system according to claim 37, wherein the first longitudinal passage extends completely through the first front bolt piece from end to end, and the second longitudinal passage extends completely through the second front bolt piece from end to end.

43. The system according to claim 36, wherein the rear bolt piece includes a firing pin.

44. A method for matching a bolt assembly to a firearm, the method comprising: providing a firearm comprising a longitudinal axis and a receiver defining an axially extending cavity; providing a barrel chambered for a first cartridge having a first propellant powder charge; providing a rear bolt piece comprising a longitudinal passage and a breech face formed in the longitudinal passage; selecting a front bolt piece from a plurality of front bolt pieces each having a different mass and a longitudinal passage, the selected front bolt piece having a mass compatible for use with firing the first cartridge in the firearm; coupling the first front bolt piece to the rear bolt piece to form a bolt assembly; and inserting the bolt assembly into the cavity of the receiver.

45. The method according to claim 44, further comprising inserting a rear breech end portion of the barrel into the cavity of the receiver and through the longitudinal passages of the selected front bolt piece and the rear bolt piece.

46. The method according to claim 44, wherein the step of inserting the bolt assembly comprises abuttingly engaging the breech face of the rear bolt piece with the rear breech end portion of the barrel.

47. The method according to claim 45, wherein the coupling step comprises inserting a coupling protrusion on the front or rear bolt pieces into a complementary configured coupling recess on the other one of the front or rear bolt pieces.

48. The method according to claim 47, wherein the coupling step includes inserting the coupling protrusion vertically upwards into the coupling recess.

49. The method according to claim 48, wherein the coupling protrusion comprises a flange and the coupling recess comprises a channel.

50. The method according to claim 48, wherein the coupling protrusion and the coupling recess are both U-shaped.

51. The method according to claim 45, wherein the step of inserting the rear breech end portion of the barrel into the cavity of the receiver further comprises rotatably engaging a threaded portion of the barrel with a threaded socket of the receiver.
